* Assignment - 5 *

Q.1 Explain file allocation methods in details.

→ File allocation is how the operating system stores files on disk blocks.

There are three main methods:

1) Contiguous Allocation:
- Files are stored in adjacent blocks.
- Fast access, simple.
- Cause external fragmentation & hard to expand files.

2) Linked Allocation:
- Each file is a linked list of blocks stored anywhere on disk.
- No fragmentation, easy to grow.
- Slow access, pointer overhead.

3) Indexed Allocation:
- Each file has an index block that stores pointers to all its data blocks.
- Supports direct access, no fragmentation.
- Requires extra space for index block.

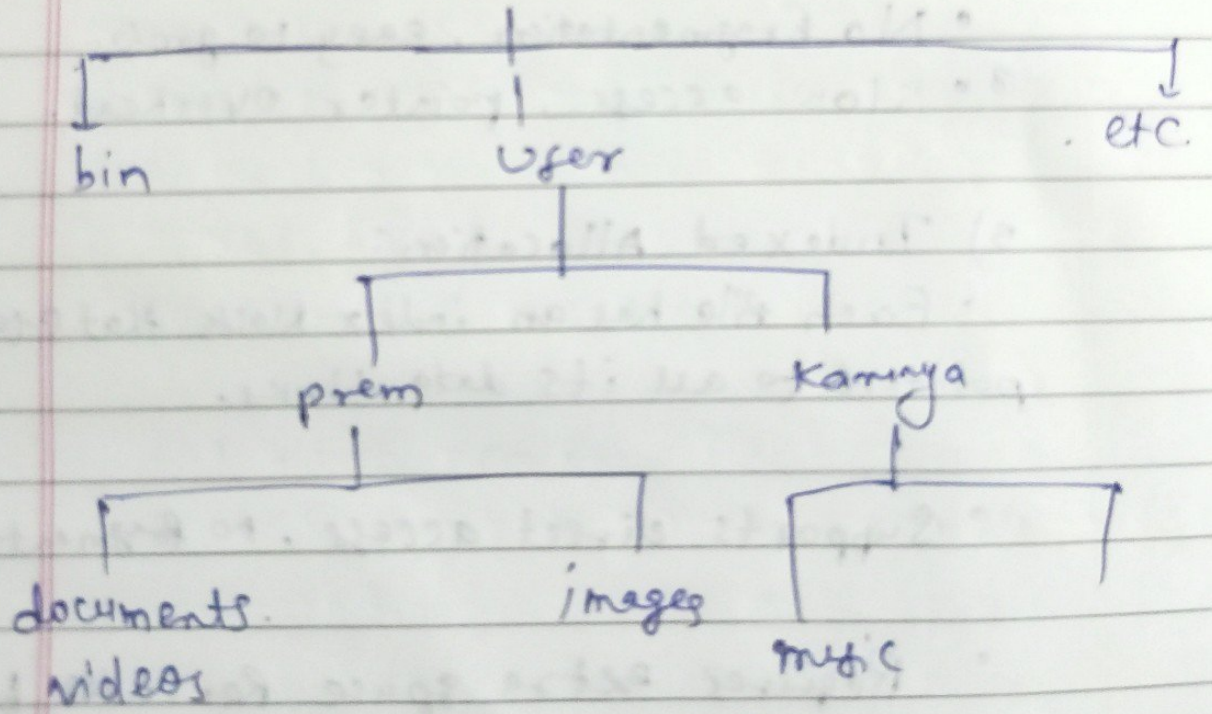
Q.2 what is tree structured directory? Explain with suitable diagram.

→ A tree-structured dictory is a type of directory organization in which directories are arranged in hierarchical (tree-like) structure. It allows users to organize files systematically into directories & subdirectories.

Each directory can contain:-

- Files
- Subdirectories (folders inside folders)

Diagram



Advantages:

- Easy to organize & locate files.
- Supports file sharing among users.

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Q.8 what are the file access methods? Explain them in detail.

→ 1) Sequential Access:

- Records are accessed one after another, from beginning to end.

- Example : Reading a log file line by line.
- Used in : Tape drives, text files.

2) Direct (Random) Access:

- Any record can be read or written directly using its address or position.

- Example : Accessing record no. 5 directly in a file.

3) Indexed Access:

- An index is created with key-record address pairs.

- Allows both sequential & direct access using index.

4) Hashed Access:

- A hashed function computes the address from the key.
- provides very fast access for exact matches.

Q.4 Describe FCFS disk scheduling policy with an example.

→ FCFS (First Come First Serve) Disk Scheduling

In the FCFS disk scheduling algorithm, the requests are served in the order they arrive in the queue - just like a regular queue.

- working
- the disk head moves from the current position to the first requested track.

- Then it moves sequentially to the next request regardless of the distance.

example :-

- Request queue = [98, 183, 37, 122, 14, 124, 65, 67]
- Initial head position = 53

Step-by-step movement :-

Head moves from :-

53 → 98 → 183 → 37 → 122 → 14 → 124 → 65 → 67.

Q.5 Explain file system in operating system.

→ A file system is a method used by an OS to store, organize, & manage data on storage devices like hard drives, SSDs, or USB drives. It helps user & program easily access, read, write, & manage files.

Function of a file system.

1) File organization:
Arrange files logically for easy access.

2) File Naming:
provides names & extensions (e.g. report.txt, image.jpg).

3) File Access control:
Manages permissions (read, write, execute).

4) File opⁿ:
Allow opⁿ like create, delete, read, write. & modify.

Types of file systems.

1) FAT (File Allocation Table) : Used in older systems & USB drives.

2) NTFS (New Technology File system):

(6) Explain SSTF & C-Scan disk scheduling algorithm with example.

→ 1) SSTF selects the disk I/O request that is closest to the current head position (i.e. has the smallest seek time).

Key idea :-

→ Always move the disk arm to the nearest track next.

→ Reduces total head movement compared to FCFS.

→ May cause starvation for requests far from the current head position.

Example :-

Let the disk queue be :-

[82, 170, 43, 140, 24, 16, 190]. & the initial head position = 50.

2) C-SCAN (Circular SCAN) Algorithm:

C-SCAN (Circular SCAN) moves the head in one direction only (say from start to end). when it reaches the last track, it jumps to the beginning without servicing requests on the way back.

key :

example :-

Same queue : [82, 190, 43, 140, 24, 16, 190]

initial head = 50

DISK range : 0 - 199.